LUNZ-ARC: A Multi-Agent Reinforcement Learning Architecture with Centralized Training and Decentralized Execution for Robust Tokamak Plasma Stabilization

Luiz Guilherme

Lunz Research Group

Working manuscript for technical review | October 2026

ABSTRACT

Maintaining plasma stability in tokamak fusion devices is a demanding feedback-control problem involving nonlinear dynamics, rapid disturbances, and strict operational constraints. This manuscript presents LUNZ-ARC, a proposed multi-agent reinforcement-learning architecture based on centralized training and decentralized execution (CTDE), with six specialized agents and an independent safety supervisor that can transfer control to a deterministic feedback fallback when a synthetic risk metric exceeds $R > 0.55$. The supplied simulation figures report evaluation under six stress profiles, including clustered pulses, sustained forcing, rising perturbation ramps, altered model parameters, and increased sensor noise. In the reported low-order synthetic environment, the hybrid RL-plus-fallback configuration maintains full episode survival across the tested profiles, while the pure-RL and uncontrolled configurations show earlier terminations in several scenarios. These results are evidence about the supplied simulation model only; they do not establish stabilization of a physical tokamak. The source materials also contain a method-description inconsistency: the text identifies MAPPO, while one supplied figure analyzes a Q-table. The implementation and figure provenance must be reconciled before algorithm-specific conclusions are treated as confirmed.

Keywords: nuclear fusion; tokamak control; multi-agent reinforcement learning; CTDE; plasma stabilization; safety-critical control; synthetic simulation

1. Introduction

Magnetically confined plasma control is a nonlinear, multivariable problem. Tokamak operation requires coordinated regulation of magnetic-field coils and other actuators to maintain plasma position, shape, and operating conditions while responding to instabilities and external disturbances. Conventional feedback methods remain important because their behavior can be analyzed and bounded, but their performance depends on controller design, model fidelity, and operating regime.

Reinforcement learning (RL) offers a complementary approach: a policy can be optimized through interaction with an environment rather than derived entirely from a fixed analytical control law. Previous work has demonstrated deep-RL-based magnetic control on the TCV tokamak, showing that learned policies can control selected plasma configurations under experimental conditions [1]. Such results motivate further investigation, but they do not remove the need for reliable training environments, validation, and safety mechanisms.

A key concern for safety-critical control is the behavior of a learned policy outside its training distribution. A policy may perform well in common states yet fail under clustered disturbances, sensor noise, or changes in plant dynamics. LUNZ-ARC addresses this concern architecturally by pairing specialized learned policies with a separate safety supervisor and a classical fallback controller. The present work focuses on the control

architecture and its reported behavior in a low-order synthetic model, rather than on a complete reactor design.

For intuition, plasma control resembles balancing a flexible pole while disturbances arrive unpredictably. A learned controller can adapt its response, but an independent safety mechanism is needed to reject or override actions when the system approaches a pre-defined risk boundary.

2. LUNZ-ARC architecture

2.1. State representation and synthetic dynamics

The supplied low-order environment tracks normalized macroscopic variables: horizontal displacement $x \in [-1, 1]$, plasma velocity $v \in [-1, 1]$, and an instability index $i \in [0, 1.25]$. An instantaneous risk metric $R(x, v, i, d) \in [0, 1]$ is computed from the state and external loading or forcing d. These variables provide a simplified test bed for control-policy evaluation; they are not a substitute for magnetohydrodynamic (MHD) simulation or diagnostic reconstruction in an experimental tokamak.

2.2. Multi-agent decomposition

LUNZ-ARC separates control responsibilities across six specialized functional agents rather than relying on a single monolithic policy. The intended roles include plasma-state observation, equilibrium and position management, instability monitoring, actuator coordination, fuel and exhaust monitoring, and joint action coordination. The exact observation and action spaces for each agent should be specified in the implementation documentation so that the architecture can be reproduced.

The multi-agent design is intended to encourage specialization while preserving cooperation. As in a coordinated pit crew, individual functions are assigned to agents, but the overall outcome depends on their joint behavior. Adding agents does not automatically improve control; the claimed benefit must be established by comparison with simpler controllers and ablation tests.

2.3. Centralized training with decentralized execution

Under the CTDE paradigm, a centralized learner can access global state information and joint actions during training. A centralized critic can estimate the value of the joint behavior, helping agents learn cooperative policies. At execution time, training-only components such as the centralized critic are removed from the control loop; each agent acts using its permitted observations and shared state information. This separation can reduce inference-time dependencies, but communication paths and timing requirements still need to be documented and measured.

The source description identifies Multi-Agent Proximal Policy Optimization (MAPPO) as the training algorithm. MAPPO is a recognized cooperative multi-agent RL baseline [2]. However, the provided policy-analysis graphic displays a Q-table and describes visited state-action cells. A tabular policy visualization is not, by itself, evidence that the plotted policy was generated by MAPPO. The algorithm, saved policy artifacts, and figure provenance therefore need to be reconciled before MAPPO-specific performance claims can be verified.

2.4. Independent safety supervisor

During execution, a supervisor monitors the synthetic risk metric and the proposed actions. When R exceeds the configured threshold of 0.55, the supervisor overrides the learned policy and invokes a deterministic classical feedback controller. This is a design rule in the simulation. It should not be described as an infallible safety guarantee: the effectiveness of the mechanism depends on risk-estimation accuracy, fallback behavior, actuator limits, timing, and coverage of failure modes.

3. Experimental setup

3.1. Stress profiles

The provided results describe six synthetic stress profiles:

- Double pulses: rapid sequential disturbances.
- Sustained forcing: continuous high-amplitude external loading.
- Rising ramp: gradually escalating perturbation load.
- Clustered pulses: groups of disturbance events arriving in bursts.
- Altered parameters: shifts in the synthetic plant dynamics.
- Increased sensor noise: higher-variance noise added to diagnostic observations.

3.2. Controller configurations and reported protocol

Four controller configurations are compared in the supplied material: pure RL, RL with the safety fallback (LUNZ-ARC), classical feedback, and no control. The text reports 12,000 training episodes and 12 paired seeds for evaluation. The plots likewise describe paired-seed comparisons. The manuscript materials supplied with this article do not include raw trajectories, a configuration file, training logs, or executable scripts sufficient to independently reproduce every plotted statistic; these artifacts should be released with a final submission.

The synthetic episode ends when a modeled boundary is crossed or the configured horizon is completed. Therefore, “survival” in this paper means survival of a simulated episode under the stated model and termination rule. It must not be interpreted as a physical probability of tokamak discharge survival.

4. Results

4.1. Episode behavior under clustered pulses

Figure 1 shows a paired episode under clustered-pulse forcing. The supplied traces compare disturbance load, displacement, velocity, instability, risk, actuator corrections, and the source of control. In this example, the pure-RL trajectory reaches the synthetic model boundary at approximately 19 s, while the hybrid configuration switches to fallback control as risk rises. This is an illustrative single episode; it does not replace aggregate statistics across all seeds.

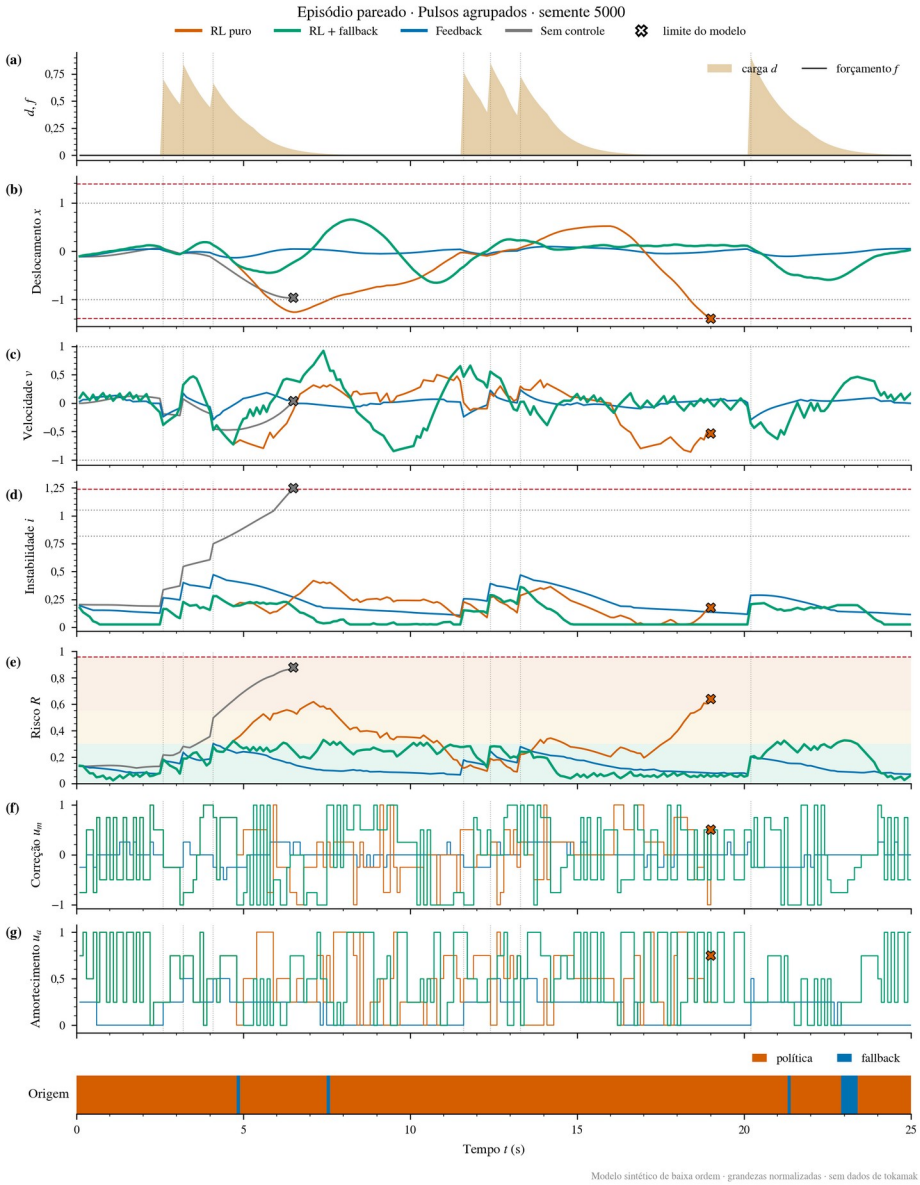


Figure 1. Example synthetic episode under clustered-pulse forcing. Panels show disturbance load, displacement, velocity, instability, risk, actuator commands, and control source.

4.2. Phase-space trajectories and risk envelopes

Figure 2 compares phase-space trajectories across the four controller configurations. In the supplied plots, uncontrolled and pure-RL trajectories more frequently approach or cross the model's indicated high-risk boundaries, while the hybrid and classical-feedback trajectories remain more tightly bounded. The result supports further testing of a fallback architecture within this synthetic environment; it does not demonstrate robustness to all possible disturbances or to real plasma dynamics.

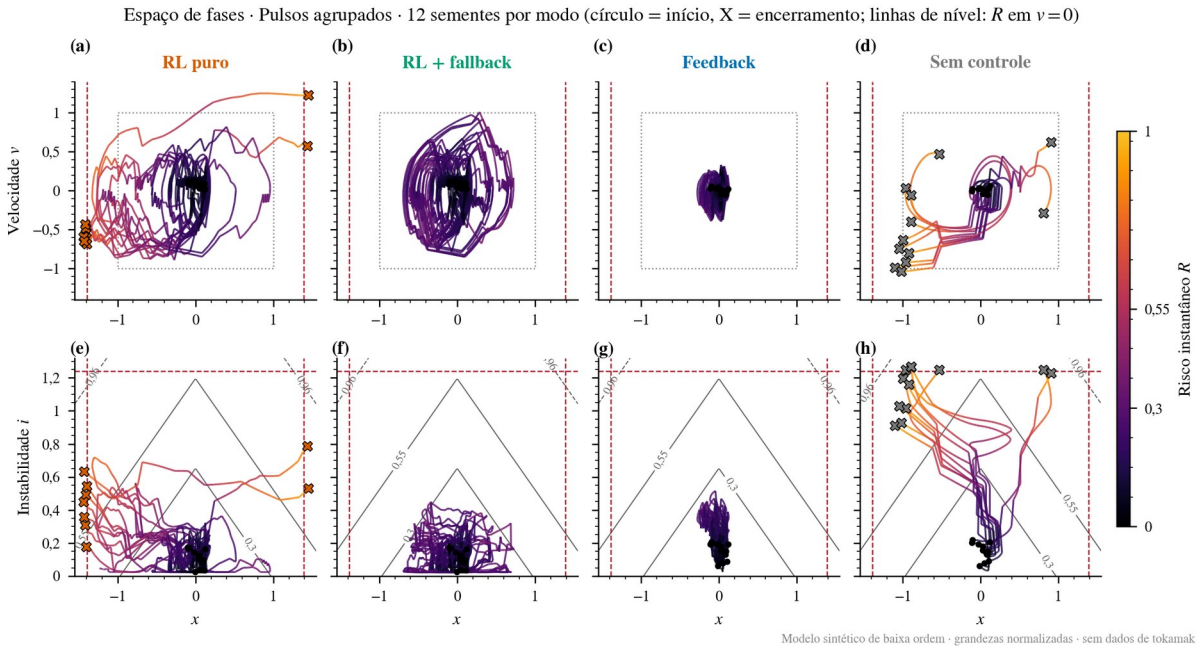


Figure 2. Synthetic phase-space trajectories under clustered-pulse forcing for pure RL, RL with fallback, classical feedback, and no control. Dashed boundaries and risk contours are model-defined.

4.3. Illustrative poloidal cross-section

Figure 3 visualizes the simulated plasma boundary and illustrative magnetic-shape changes for the controller configurations. The hybrid and classical-feedback rows appear more centered in the displayed sequence, while the pure-RL and uncontrolled rows show larger deformation before model termination. These panels are visualizations of the low-order model, not evidence from a full MHD solver or an experimental tokamak.



Figure 3. Simulated poloidal cross-section across selected times in a clustered-pulse episode. This is an illustrative view of the synthetic model, not a full MHD reconstruction.

4.4. Generalization across stress profiles

Figure 4 summarizes the supplied benchmark across six perturbation profiles. Its panels report early-termination fraction, mean episode risk, and the fraction of steps above the model risk threshold. The supplied plots indicate that the hybrid configuration avoids early termination in the evaluated paired seeds and achieves risk measures comparable to the classical-feedback baseline. Exact confidence intervals and effect sizes should be interpreted alongside the raw data and should be regenerated from the released benchmark scripts before formal publication.

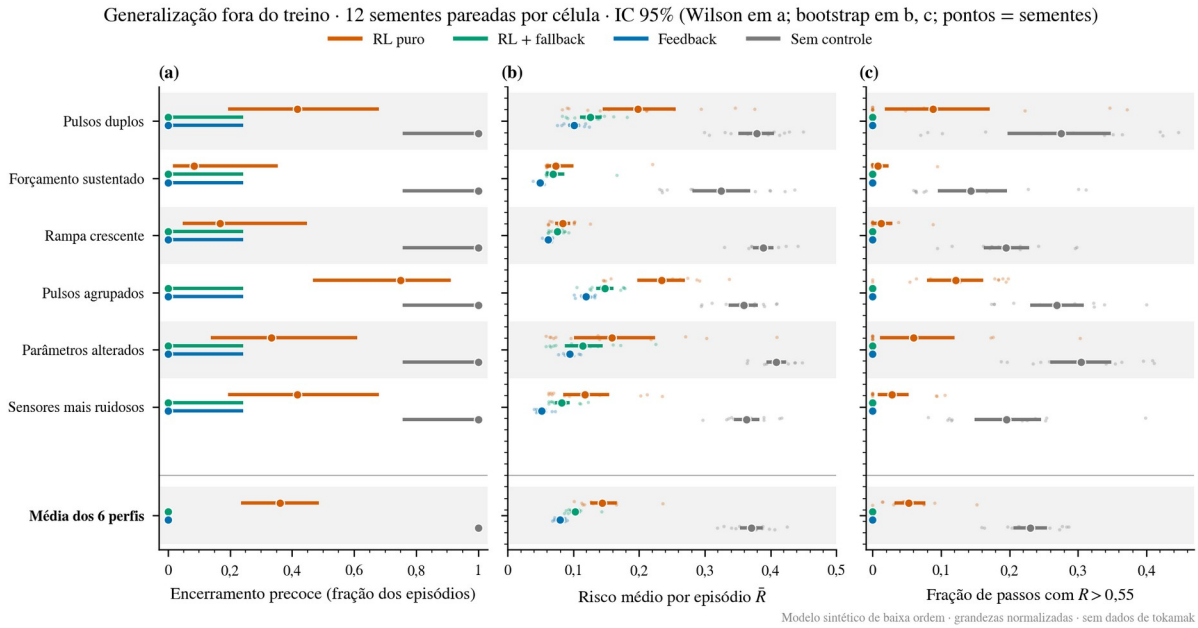


Figure 4. Benchmark summary for six synthetic stress profiles. The three panels report early termination, mean modeled risk, and the fraction of steps above the risk threshold.

4.5. Survival analysis

Figure 5 presents Kaplan–Meier estimates of simulated episode survival with Greenwood 95% confidence bands. As reported in the supplied material, the uncontrolled and pure-RL configurations show reductions in survival under several stress conditions, whereas the RL-plus-fallback and classical-feedback curves remain at or near 1.0 in the tested runs. These results are conditional on the model, threshold, seed set, and termination rule. A flat curve in this benchmark is not a general proof of safety, nor does it establish a zero failure probability outside the tested scenarios.

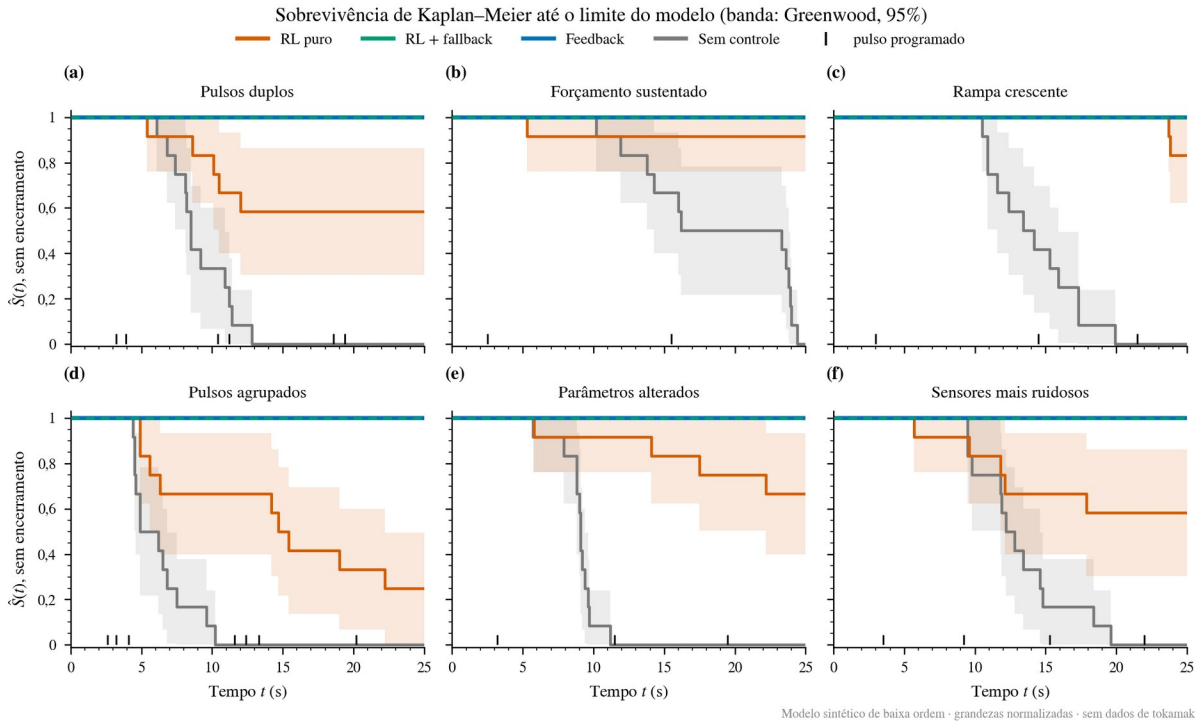


Figure 5. Kaplan–Meier estimates of synthetic episode survival across the six stress profiles, with the confidence bands shown in the supplied plot.

4.6. Policy and action analysis

The supplied Q-table graphic (Figure 6) reports that 62% of 6,435 visited state-space cells converged to Action 0, identified in the source text as u_m = −1 and u_a = 0. The source interprets this as reduced unnecessary actuator activity under nominal conditions, with more frequent corrections under strong disturbances. Before drawing firm conclusions, the state discretization, visit counts, policy-selection rule, and relationship between this table and the stated MAPPO training pipeline must be documented. This is the primary implementation-level inconsistency identified in the supplied materials.

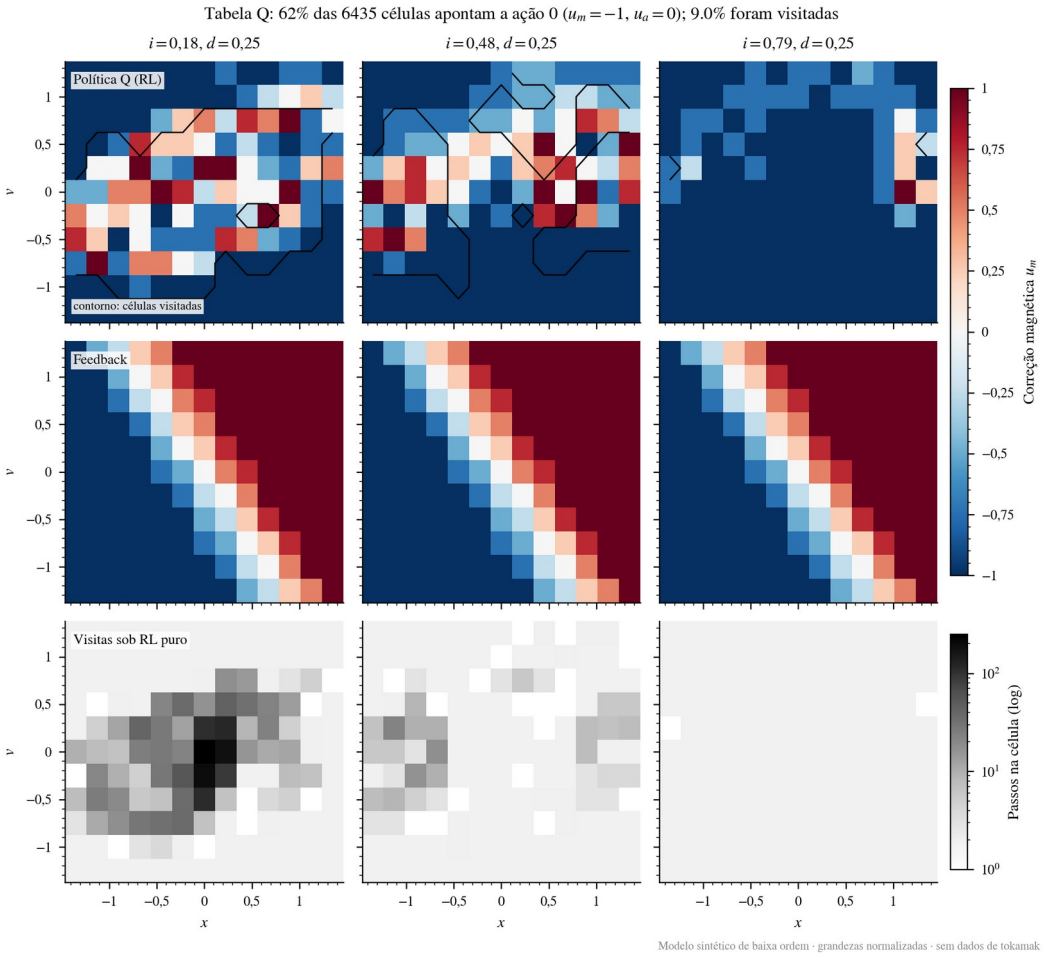


Figure 6. Supplied policy-analysis visualization, including Q-table/action maps and state-visitation summaries. The source figure needs to be reconciled with the MAPPO description in the methods.

4.7. Example simulator view

Figure 7 provides an example visual snapshot of the simulator under a noisy-sensor scenario. It is useful for communicating the state visualization and displayed time-series, but the screenshot alone does not provide a reproducible quantitative result.

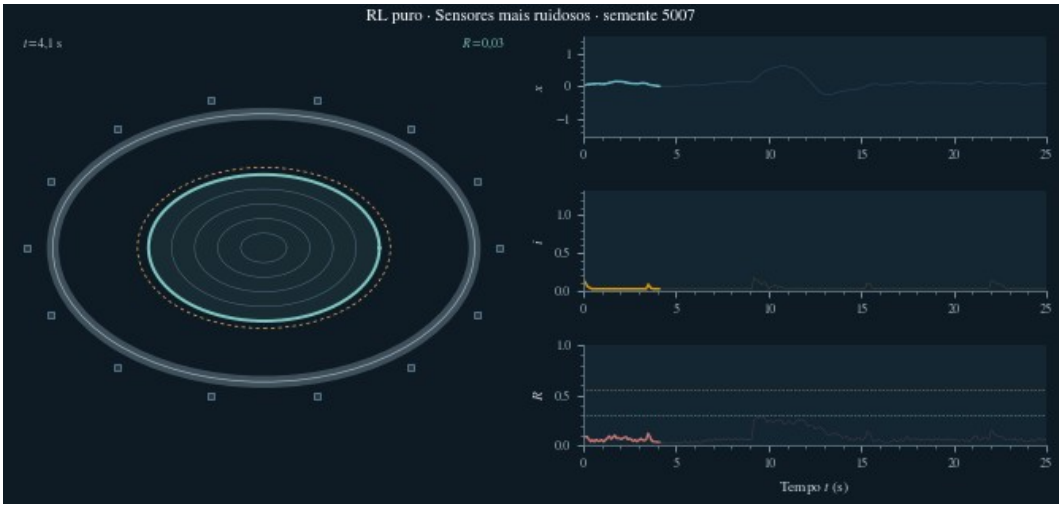


Figure 7. Example simulator view for pure RL under the increased-sensor-noise profile (seed 5007), as supplied by the author.

5. Discussion

The supplied results support a narrow conclusion: in the configured low-order synthetic environment, a deterministic fallback triggered by a modeled risk threshold appears to reduce episode terminations relative to the pure-RL and no-control configurations. The classical-feedback controller also performs strongly in the reported metrics. The results therefore motivate investigation of hybrid control, rather than establishing that a multi-agent RL controller is superior to classical feedback.

Three questions remain central. First, does the learned policy add value when compared fairly with classical feedback, including control effort, tracking accuracy, and recovery time? Second, are the reported gains robust to model uncertainty and to disturbance distributions not represented during training? Third, can the risk supervisor detect hazardous states early enough when observations are noisy, delayed, or incomplete? These questions require ablation tests and broader stress testing.

The architecture also needs more complete implementation documentation. In addition to resolving the MAPPO/Q-table mismatch, a reproducible study should make available the source code, environment equations, reward terms, training seeds, hyperparameters, termination conditions, risk calculation, and raw per-episode records. Reporting mean values without the underlying data limits independent scrutiny.

6. Limitations and next steps

This work is limited to a synthetic low-order model with normalized states. It does not capture the full MHD dynamics, diagnostic complexity, actuator bandwidth, material response, or engineering constraints of a physical tokamak. The risk threshold $R > 0.55$ is a model parameter, not a validated operational limit. The results should not be used to control an experimental device.

The next stages are to (i) reconcile the learning algorithm and policy artifacts; (ii) publish the environment, training code, model checkpoint, and raw benchmark data; (iii) rerun the evaluation with predefined statistical procedures and a larger set of seeds; (iv) test ablations of the fallback supervisor and agent decomposition; and (v) progress toward higher-fidelity MHD environments only after the lower-order model is independently reproducible. Codes such as JOREK and M3D-C1 are examples of established MHD simulation tools relevant to future research [3, 4]. Hardware-in-the-loop work should follow only after specialist safety review and independent validation.

7. Conclusion

LUNZ-ARC proposes a hybrid control architecture that combines specialized learning-based policies with an independent classical fallback under a CTDE training framework. The supplied synthetic results indicate improved episode survival for the RL-plus-fallback configuration compared with pure RL and no control in the evaluated scenarios, while classical feedback remains a strong baseline. The contribution should therefore be framed as a research hypothesis and an initial simulation study, not as validation of robust real-tokamak stabilization. Resolving the algorithm/figure discrepancy and publishing reproducible artifacts are prerequisites for a defensible external review.

References

1. Degrave, J., Felici, F., Buchli, J., et al. “Magnetic control of tokamak plasmas through deep reinforcement learning.” Nature 602, 414–419 (2022). https://doi.org/10.1038/s41586-021-04301-9

2. Yu, C., Velu, A., Vinitsky, E., et al. “The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games.” Advances in Neural Information Processing Systems 35 (NeurIPS 2022).

3. Hoelzl, M., Huijsmans, G. T. A., Artola, F. J., et al. “Non-linear MHD modelling of transients in tokamaks: a review of recent advances with the JOREK code.” Nuclear Fusion 64, 112016 (2024). https://doi.org/10.1088/1741-4326/ad5a21

4. Princeton Plasma Physics Laboratory. “M3D-C1: Overview.” https://m3dc1.pppl.gov/ (accessed October 2026).